

ABOUT ME

Software engineer and researcher working on Quantum Computing. Experienced in developing full-stack applications, optimizing machine learning models and quantum computing algorithms. Background in containerization, and automation for AI-driven applications in Python and Rust. Passionate about bridging research and software development to create efficient, secure, and innovative solutions.

EDUCATION

Institute of Science Tokyo (Former Tokyo Institute of Technology) Ph.D. in Information and Communication Engineering (<i>Super Smart Society Incentive</i>)	<i>Sept. 2023-Present</i>
The High Energy Accelerator Research Organization (KEK) M.Sc. in Experimental High Energy Physics (<i>Monbukagakusho Honors</i>)	<i>Oct. 2020-Sept. 2023</i>
University of California, San Diego B.Sc. in Physics (<i>Provost Honors</i>)	<i>June 2016-Mar. 2020</i>

TECHNICAL SKILLS

Programming: Python, C++, Rust, Typescript, Kotlin, Julia
Packages or Libraries: PyTorch, JAX, PennyLane, Qiskit, Manifolds.jl/Manopt.jl, Scikit-learn, Flask, React, GEANT4
Dev Tools: Docker, Grafana, MongoDB, Portainer, PostgreSQL
Languages: English (Professional), Japanese (Business Intermediate), Mandarin (Native)

EXPERIENCE

Research Associate RIKEN Advanced Intelligence Project Developed data extraction and membership-inference attacks and defenses for large language and vision-language models, plus RL-based alignment pipelines. Conducted theoretical research on AI security and privacy.	<i>Apr. 2024-Sept. 2026</i>
Visiting Scholar HKUST-Guangzhou Developing package with rigorous mathematical foundation and scientific computing tools in quantum computing, and provide issues for problem reduction in complexity theory.	<i>Jan. 2026-Mar. 2026</i>
Software Engineer Intern ExaWizards Inc. Developed a BtoB web application for enterprise HR skill training using React and Kotlin. Implemented database migration and data flow design to enhance system efficiency and user experience.	<i>Aug. 2025-Dec. 2025</i>
Junior Physicist Inner Tracking Detector Group (ATLAS Exp., CERN) Analyzed Higgs self-coupling physics ($HH \rightarrow b\bar{b}\tau\tau$) with custom CNN/GNN pipelines on ATLAS pixel-detector data. Collaborated with CERN on detector testing and data interpretation.	<i>Apr. 2021-Sept. 2023</i>
Lab Assistant Popmintchev Lab, UCSD Designed optical systems for high harmonic generation. Simulated intensity profiles of laser beams using Matlab for ionized gases (Ar^+ , He^+ , Ne^+).	<i>July 2018-Mar. 2020</i>

PROJECTS

pdfT: Trainable Unitary Bases for Image Compression and Inpainting JAX library of trainable sparse bases (QFT, entangled QFT, TEBD, MERA, DCT-IV) optimized on the unitary manifold so the transform stays exactly invertible; the code behind arXiv:2608.00053 and arXiv:2609.17298. Supersedes the Julia original ParametricDFT.jl. lib bench <i>Python, JAX, Julia, Manopt.jl</i>	<i>Dec. 2025-Present</i>
problem-reductions: Extensible Framework for Computational Complexity Rust library and pred CLI cataloguing 100+ NP-hard problems and 200+ reductions, routing any problem to a registered solver through the reduction graph; developed with a human-agent engineering pipeline (arXiv:2604.11535). lib <i>Rust, Claude Code</i>	<i>Jan. 2026-Apr. 2026</i>
LogosQ: Quantum Computing Library in Rust Type-safe simulator with gates, circuits, and optimized direct-gate and tensor-contraction backends (arXiv:2512.23183). lib doc <i>Rust, TypeScript, Docker</i>	<i>July 2025-Present</i>
ManifoldsGPU.jl: GPU-Accelerated Riemannian Optimization CUDA acceleration for the JuliaManifolds ecosystem: batched kernels for manifold operations and retractions, about 20× faster exp! on PowerManifold(Stiefel(32,16), 2048) and 15× for batched-SVD polar retraction. lib <i>Julia, CUDA.jl, Manifolds.jl</i>	<i>Jan. 2026-Present</i>

PUBLICATIONS

Quantum-Inspired Trainable and Parameter-Efficient Tensor Networks for Image Inpainting

S. An, K. Slavakis. arXiv:2609.17298, 2026. Submitted to ICASSP 2027. [code]

Fast Trainable Multilinear Bases for Image Compression

S. An, Z. Ni, H. Zhou, J.-G. Liu. arXiv:2608.00053, 2026. Submitted to *Quantum*. [code]

Problem Reductions at Scale: Agentic Integration of Computationally Hard Problems

X.-W. Pan, S. An, J.-G. Liu. arXiv:2604.11535, 2026. Submitted to *NeurIPS* 2026. [code]

LogosQ: A High-Performance and Type-Safe Quantum Computing Library in Rust

S. An, J. Wang, K. Slavakis. arXiv:2512.23183, 2025. Under review at *IEEE Transactions on Quantum Engineering*. [code]

Tensor-Based Binary Graph Encoding for Variational Quantum Classifiers

S. An, K. Slavakis. *IEEE QCNC*, Nara, Japan, 2025. arXiv:2501.14185

Combination of Searches for Heavy Spin-1 Resonances Using 139 fb^{-1} of Proton-Proton Collision Data at $\sqrt{s} = 13\text{ TeV}$ with the ATLAS Detector

ATLAS Collaboration. *Journal of High Energy Physics* 2024, 118, 2024. doi:10.1007/JHEP04(2024)118

Search for Nearly Mass-Degenerate Higgsinos Using Low-Momentum Mildly Displaced Tracks in pp Collisions at $\sqrt{s} = 13\text{ TeV}$ with the ATLAS Detector

ATLAS Collaboration. *Physical Review Letters* 132, 221801, 2024. doi:10.1103/PhysRevLett.132.221801

Overnight Stock Movement Prediction with Contextualized Embedding Incorporating News and Stock

J. H. Xu, S. An, L. Lyu, Y. Bai. *IEEE 11th Global Conference on Consumer Electronics (GCCE)*, pp. 476–478, 2022. doi:10.1109/GCCE56475.2022.10014070

AWARDS AND HONORS

Science Tokyo Spring Scholarship — next-generation front-runner development program.	2025.12-Now
TSUBAME Scholarship — for outstanding Ph.D. students at Institute of Science Tokyo.	2025.10-12
Super Smart Society Research Incentive — fundamental technologies for a Super Smart Society.	2024.4-Now
JASSO Monbukagakusho Honors Scholarship — government award for graduate study at KEK.	2021-2022
Provost Honors — UCSD Provost's Honor List for undergraduate academic performance.	2016-2020

ADDITIONAL ACTIVITIES

- Contribute to multiple open source projects on GitHub, including PennyLane v0.40.0 and JuliaManifolds.
- Board Game Arena, Catan Global Top 50. + Scythe, Brass Birmingham.
- PADI Advanced Open Water Diver (Licensed 2018).
- GO, Amateur 2 Dan.